Formula List for Diffraction

1. Grating formula

$$(a+b)\sin\theta = n\,\lambda$$

Where;

- $(a+b)$= Grating element
- θ = Angle of Diffraction
- n = order of Diffraction
- λ = Wavelength of wave getting Diffracted

2. Grating Element

$$(a+b) = \frac{1}{N}$$

Where;

- $(a+b)$= Grating element
- N = Number of lines per unit length of Grating

3. Resolving power of Grating (R.P.)

$$\mathrm{R.P.} = \frac{\lambda}{\mathrm{d}\lambda} = mN$$

Where;

- N = Number of lines per unit length of Grating
- m = order of Diffraction
- λ = Mean Wavelength of sources to be resolved
- d λ = Difference between Wavelength of sources to be resolved